



M.Tech Project Proposal

Programme: M.Tech (Artificial Intelligence & Machine Learning)

Institute: *Indian Institute of Technology Jodhpur*

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Project Duration: Two Semesters (Project-I + Project-II)

Credits: 5 (Semester I) + 11 (Semester II)

1. Title of the Project

Ontology-Guided Multilingual Semantic Retrieval and Explainable AI Framework for Classical Ayurvedic Knowledge Systems

2. Introduction and Motivation

Ayurveda is a traditional Indian medical system with a history spanning several millennia. Its foundational knowledge is preserved in classical Sanskrit texts such as *Charaka Samhita*, *Sushruta Samhita*, and *Ashtanga Hridaya*. These texts contain detailed descriptions of diseases, symptoms, treatments, and herbal formulations, forming a rich but complex knowledge base.

Despite their importance, Ayurvedic texts remain largely inaccessible to modern computational systems due to challenges such as classical Sanskrit language usage, lack of structured digital representations, limited availability of machine-readable datasets, and the absence of explainable AI frameworks tailored to this domain. Existing search engines and AI-based systems either rely on keyword-based retrieval or unconstrained generative models, which are inadequate for accurate, explainable, and verifiable knowledge access—especially in medical contexts.

Recent advancements in Artificial Intelligence and Machine Learning, particularly in multilingual natural language processing (NLP), semantic search, knowledge graphs, and retrieval-augmented generation (RAG), provide an opportunity to systematically address these challenges. This project aims to design a principled AI framework that enables semantic retrieval and explainable reasoning over classical Ayurvedic texts while preserving Sanskrit as the authoritative source language.

3. Problem Statement

The central problem addressed in this project is the lack of an intelligent, explainable, and multilingual retrieval system for classical Ayurvedic knowledge.

Specifically, the project seeks to answer the following research question:

How can multilingual semantic retrieval be combined with ontology-guided reasoning to build an explainable and faithful AI system for classical Ayurvedic texts written in Sanskrit?

4. Objectives

The key objectives of this project are:

1. To create a clean, structured, and machine-readable corpus from classical Ayurvedic texts.
2. To develop a multilingual semantic retrieval system capable of handling Sanskrit, Hindi, and English queries.
3. To design and implement an ontology and knowledge graph representing core Ayurvedic concepts.
4. To integrate ontology-guided reasoning with neural retrieval models.
5. To generate explainable and citation-backed responses using retrieval-augmented generation.
6. To rigorously evaluate the proposed system using standard information retrieval metrics.

5. Scope of the Work

- Primary focus on *Charaka Samhita* and *Ashtanga Hridaya*.
- Sanskrit texts will be treated as the authoritative knowledge source.
- Hindi and English translations will be used as aligned explanatory layers.
- The project focuses on research and experimentation rather than full-scale clinical deployment or UI development.

6. Proposed Methodology

6.1 Data Acquisition and Preprocessing

- Collection of digital and scanned versions of classical texts.

- OCR for scanned documents and normalization of Sanskrit text.
- Semantic chunking with metadata such as chapter, verse, and topic.

6.2 Multilingual Semantic Retrieval

- Generation of sentence-level embeddings using multilingual transformer models.
- Implementation of vector-based semantic search.
- Keyword-based retrieval (BM25) as a baseline.

6.3 Ontology and Knowledge Graph Construction

- Manual and semi-automatic construction of an Ayurvedic ontology.
- Representation of entities such as diseases, symptoms, doshas, herbs, and treatments.
- Storage and querying using a graph database.

6.4 Explainable Retrieval-Augmented Generation

- Integration of retrieved text with a language model to generate grounded responses.
- Explicit citation of source texts (grantha, chapter, verse).
- Preservation of Sanskrit terminology with optional explanations.

6.5 Evaluation

- Quantitative evaluation using Precision@K, Recall@K, MAP, and NDCG.
- Comparative analysis between keyword-based, semantic-only, and ontology-guided retrieval.
- Qualitative evaluation of explainability and faithfulness.

7. Semester-wise Work Plan

Semester I – Project-I (5 Credits)

- Detailed literature review
- Dataset collection and digitization
- Text preprocessing and semantic chunking
- Implementation of baseline multilingual semantic retrieval
- Preliminary evaluation and progress report

Deliverables:

- Clean Ayurvedic text corpus
- Baseline retrieval system
- Project-I report

Semester II – Project-II (11 Credits)

- Ontology and knowledge graph development
- Ontology-guided hybrid retrieval
- Integration of explainable RAG framework
- Extensive experimental evaluation and ablation studies
- Thesis writing and paper submission

Deliverables:

- Complete AI system prototype
- M.Tech thesis

- Research paper submission to an international venue

8. Expected Outcomes

- A novel ontology-guided semantic retrieval framework for Ayurvedic texts
- A multilingual, explainable AI system with citation-backed outputs
- A reusable dataset and ontology for future research
- Publications suitable for international journals or conferences

9. Research and PhD Continuation Potential

The proposed work offers strong scope for doctoral research in areas such as:

- Neuro-symbolic AI and hybrid reasoning
- Knowledge graph learning and graph neural networks
- Evaluation of faithfulness and explainability in medical AI
- Extension to other traditional knowledge systems

10. Conclusion

This project combines modern AI and ML techniques with classical Indian knowledge systems to address a challenging and impactful research problem. By emphasizing semantic retrieval, ontology-driven reasoning, and explainability, the proposed work aligns well with the academic rigor expected in an M.Tech program at a premier Indian institute and provides a strong foundation for continued doctoral research.